

Topic I SOLUTION

Given: $N=6$; $x_1=0.0557$; $x_2=1.8529$; $x_3=1.9311$; $x_4=2.1905$; $x_5=3.8884$; $x_6=4.4384$.

1) $\mu_{(ML)} = ?$

$$\mu_{(ML)} = \frac{1}{N} \sum_{n=1}^N x_n = \boxed{2.3928}$$

2) $\sigma_{(ML)}^2 = ?$

$$\sigma_{ML}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{(ML)})^2 = \boxed{2.0715}$$

3) $\mu_{(ML)}$ is Unbiased.
 σ_{ML}^2 is biased.

4) Unbiased est. of σ_{ML}^2 is : $\frac{1}{N-1} \sum_{n=1}^N (x_n - \mu_{(ML)})^2$.

Putting the values of x_n, N & μ_{ML} we get

$$\sigma_{(Unbiased)}^2 = \boxed{2.4858}$$

5) $\mu_{(Bayes)} = ?$

$\sigma^2 = 1$; $\underbrace{\sigma_0^2 = 3; \mu_0 = 4}_{\text{Prior dist. parameters}}$

The estimated mean $\mu_{(Bayes)}$ will be taken as the mean of posterior dist. that is given by

$$p(u|x) = N(u | \mu_N, \sigma_N^2)$$

$$\mu_{Bayes} = \mu_N = \frac{\sigma^2}{N\sigma_0^2 + \sigma^2} \mu_0 + \frac{N\sigma_0^2}{N\sigma_0^2 + \sigma^2} \mu_{(ML)}$$

$$= \frac{1(4)}{6(3)+1} + \frac{6(3)}{6(3)+1} \cdot 2.3928 = \boxed{2.4774}$$

84. 6) a) For smaller values of N , the Bayesian estimate is expected to perform well as it takes into consideration prior information about the data distribution (in the form of prior dist). ML estimates are generally suboptimal for smaller N .

b) For $N \rightarrow \infty$, it can be shown that ML estimates become optimal as do the Bayesian ones.

7) ML only considers given data while estimation and thus less data means that estimates could be erroneous. Concretely

$$\text{Var}[\mu_{ML}] = \frac{\text{Var}[x]}{N}$$

Thus, larger the N , more accurate the estimate of ML.

Week 3) Book exercise 2.1)

Assume $x \sim \text{Bern}(x/\mu)$, i.e.,

$$p(x) = \mu^x (1-\mu)^{1-x}$$

for $x \in \{0, 1\}$.

We immediately have that also denoted $p(x/\mu)$.

$$\sum_{x \in \{0, 1\}} p(x/\mu) = \mu^0 (1-\mu)^{1-0} + \mu^1 (1-\mu)^{1-1} = 1 - \mu + \mu = 1,$$

$$E[x] \stackrel{\text{def.}}{=} \sum_{x \in \{0, 1\}} x p(x) = 0 \cdot (\mu^0 (1-\mu)^{1-0}) + 1 \cdot (\mu^1 (1-\mu)^{1-1}) = \mu,$$

and

$$\text{Var}(x) \stackrel{\text{def.}}{=} E[(x - E[x])^2]$$

$$\stackrel{\text{def.}}{=} \sum_{x \in \{0, 1\}} (x - \mu)^2 p(x)$$

$$= (0 - \mu)^2 \cdot \mu^0 (1-\mu)^{1-0} + (1 - \mu)^2 \cdot \mu^1 (1-\mu)^{1-1}$$

$$= (0 - \mu)^2 (1 - \mu) + (1 - \mu)^2 \cdot \mu$$

$$= \mu^2 - \mu^3 + (1 + \mu^2 - 2\mu) \cdot \mu$$

$$= \mu^2 - \mu^3 + \mu + \mu^3 - 2\mu^2$$

$$= \mu - \mu^2$$

$$= \mu (1 - \mu)$$

Book exercise 2.5)

Before we start, we recall integration by substitution

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

where $u = g(x)$.

Now, the exercise simply asks us to show

$$\int_0^1 \mu^{a-1} (1-\mu)^{b-1} d\mu = \frac{\Gamma(a) \Gamma(b)}{\Gamma(a+b)}$$

given that

$$\Gamma(a) + \Gamma(b) = \int_0^\infty \exp(-x) x^{a-1} dx \cdot \int_0^\infty \exp(-y) y^{b-1} dy.$$

To this end, we note that

$$\begin{aligned} \Gamma(a) \Gamma(b) &= \int_0^\infty x^{a-1} \exp(-x) dx \int_0^\infty \exp(-y) y^{b-1} dy \\ &= \int_0^\infty x^{a-1} \cdot \left(\int_0^\infty \exp(-x) \exp(-y) y^{b-1} dy \right) dx \\ &= \int_0^\infty x^{a-1} \cdot \underbrace{\left(\int_0^\infty \exp(-(x+y)) y^{b-1} dy \right)}_{(*)} dy \end{aligned}$$

Now, let us consider the integral in (*), doing integration by substitution with

$$t = g(y) = x + y, \quad g'(y) = 1$$

where we consider x to be fixed, i.e., a constant. Doing so, we get

$$\begin{aligned} (*) &= \int_0^{\infty} \exp(-(x+y)) y^{b-1} dy \\ &= \int_0^{\infty} \exp(-(x+y)) (x+y-x)^{b-1} dy \\ &= \int_x^{\infty} \exp(-t) (t-x)^{b-1} dt \end{aligned}$$

Using integration by substitution.

Formally, we have

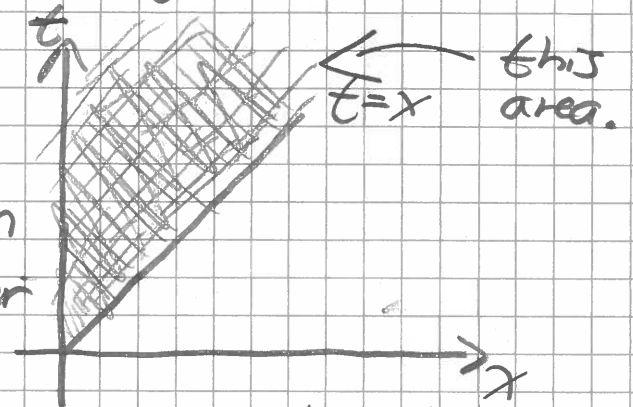
$$f(z) = \exp(-z) \cdot (z-x)^{b-1}.$$

Using our new-found expression for (*), we have

$$\Gamma(a) \Gamma(b) = \int_0^{\infty} x^{a-1} \cdot \left(\int_x^{\infty} \exp(-t) (t-x)^{b-1} dt \right) dx$$

As the exercise suggests, we would now like to interchange the order of x and t integration. But (!), we have to be careful as the integral range depends on " x ".

In the "outer" integral, we let x run from zero to ∞ . In the "inner" integral, t runs from x to ∞ . In the 2-dimensional plane, our area of integration can then be illustrated as



To get the same area of integration when we interchange the order of integration, we let t run from zero to ∞ in the "outer" integral and x run from zero to t in the "inner" integral.

We then have

$$\begin{aligned}
 \Gamma(a) \Gamma(b) &= \int_0^{\infty} x^{a-1} \left(\int_x^{\infty} \exp(-t) (t-x)^{b-1} dt \right) dx \\
 &= \int_0^{\infty} \int_x^{\infty} x^{a-1} \exp(-t) (t-x)^{b-1} dt dx \\
 &= \int_0^{\infty} \int_0^t x^{a-1} \exp(-t) (t-x)^{b-1} dx dt \\
 &= \int_0^{\infty} \exp(-t) \int_0^t x^{a-1} (t-x)^{b-1} dx dt \\
 &= \int_0^{\infty} \exp(-t) \underbrace{\left(\int_0^t \left(t \cdot \frac{x}{t} \right)^{a-1} \cdot \left(t \cdot \left(1 - \frac{x}{t} \right) \right)^{b-1} dx \right)}_{(**)} dt
 \end{aligned}$$

Now, let us consider $(**)$ (where t is fixed, i.e., treated as a constant) and do integration by substitution with

$$\mu = g(x) = \frac{x}{t}, \quad g'(x) = \frac{1}{t}.$$

Doing so, we get

$$\begin{aligned} (**) &= \int_0^t \left(t \cdot \frac{x}{t}\right)^{a-1} \cdot \left(t \cdot \left(1 - \frac{x}{t}\right)\right)^{b-1} \cdot t \cdot \frac{1}{t} dx \\ &\rightarrow \int_0^1 (t \cdot \mu)^{a-1} \cdot (t \cdot (1 - \mu))^{b-1} \cdot t d\mu \end{aligned}$$

Using integration

by substitution.

$$\text{Formally, } f(z) = (t \cdot z)^{a-1} \cdot (t \cdot (1 - z))^{b-1} \cdot t$$

$$= \int_0^1 t^{a-1} \mu^{a-1} \cdot t^{b-1} (1 - \mu)^{b-1} \cdot t d\mu$$

Using our new-found expression for $(**)$, we have

$$\begin{aligned} T(a) T(b) &= \int_0^\infty \exp(-t) \left(\int_0^1 t^{a-1} \mu^{a-1} \cdot t^{b-1} (1 - \mu)^{b-1} \cdot t d\mu \right) dt \\ &= \int_0^\infty \exp(-t) t^{a-1} t^{b-1} t dt \int_0^1 \mu^{a-1} (1 - \mu)^{b-1} d\mu \end{aligned}$$

$$= \int_0^{\infty} \exp(-t) t^{(a+b-1)} dt \int_0^1 \mu^{a-1} (1-\mu)^{b-1} d\mu$$

$$= \Gamma(a+b) \int_0^1 \mu^{a-1} (1-\mu)^{b-1} d\mu.$$

It now immediately follows that

$$\int_0^1 \mu^{a-1} (1-\mu)^{b-1} d\mu = \frac{\Gamma(a) \Gamma(b)}{\Gamma(a+b)}$$

and we are done.

Book exercise 2.34)

We want to show that

$$\Sigma_1 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu)(x_n - \mu)^T$$

maximizes (2.118), i.e., the log likelihood function of the multi-dimensional Gaussian distribution

$$\ln(p(X/\mu, \Sigma)) = -\frac{N \cdot D}{2} \ln(2\pi) - \frac{N}{2} \ln(|\Sigma|) - \frac{1}{2} \sum_{n=1}^N (x_n - \mu)^T \Sigma^{-1} (x_n - \mu)$$

where N denotes our number of observations and D denotes the number of dimensions.

Useful rule from linear algebra: let A be a quadratic $D \times D$ matrix and v_n be D -dimensional vectors for $n=1, \dots, N$. Then

$$\sum_{n=1}^N v_n^T A v_n = \text{tr}(A Q) \quad \text{where } Q = \sum_{n=1}^N v_n v_n^T.$$

- To convince yourself that the result holds, you may write out the matrix product of the left-hand side.

Now, if we let $S' = \frac{1}{N} \sum_{n=1}^N (x_n - \mu)(x_n - \mu)^T$

we have that

$$\ln(p(x/\mu, \Sigma)) = -\frac{N \cdot D}{2} \ln(2\pi) - \frac{N}{2} \ln(|\Sigma|) - \frac{N}{2} \text{tr}(\Sigma^{-1} S')$$

with

$$\frac{\partial}{\partial \Sigma} \left(-\frac{N \cdot D}{2} \ln(2\pi) \right) = 0,$$

$$\frac{\partial}{\partial \Sigma} \left(-\frac{N}{2} \ln(|\Sigma|) \right) \stackrel{(C.28)}{=} -\frac{N}{2} \left((\Sigma^{-1})^T \right) \stackrel{\text{symmetry}}{=} -\frac{N}{2} \Sigma^{-1}$$

and

$$\frac{\partial}{\partial \Sigma} \left(-\frac{N}{2} \text{tr}(\Sigma^{-1} S') \right) = \frac{N}{2} \Sigma^{-1} S' \Sigma^{-1}$$

where we have used the following rule from "The Matrix Cookbook" by Petersen & Petersen (I encourage you to download it!)

$$\frac{\partial}{\partial X} \text{tr}(A X^{-1} B) = -(X^{-1} B A X^{-1})^T = -(X^{-1})^T A^T B^T (X^{-1})^T$$

- In our case $A=I$, $X=\Sigma$, and $B=S'$. Note also that Σ and S' are symmetric.

We now have that

$$\frac{d}{d\underline{\Sigma}} (\ln(p(\underline{X}|\underline{\mu}, \underline{\Sigma}))) = 0$$

$$\Rightarrow -\frac{N}{2} \underline{\Sigma}^{-1} + \frac{N}{2} \underline{\Sigma}^{-1} \underline{S} \underline{\Sigma}^{-1} = 0$$

which directly implies that $\underline{S} = \underline{\Sigma}$.

Book exercise 2.35)

We consider a multidimensional Gaussian variable $\underline{x} \sim \mathcal{N}_D(\mu, \Sigma)$.

We are first asked to show that

$$\mathbb{E}[\underline{x}\underline{x}^T] = \mu\mu^T + \Sigma.$$

- There is actually a direct proof of this in the book. Else, we may note that it is simply a consequence of the multidimensional version of a relation we showed in week 1,

$$\begin{aligned}\Sigma &= \text{Cov}(\underline{x}, \underline{x}) = \mathbb{E}[\underline{x}\underline{x}^T] - \mathbb{E}[\underline{x}]\mathbb{E}[\underline{x}]^T \\ &\stackrel{(2.59)}{=} \mathbb{E}[\underline{x}\underline{x}^T] - \mu\mu^T\end{aligned}$$

We are now asked to show that

$$\mathbb{E}[\underline{x}_n \underline{x}_n^T] = \mu\mu^T + \mathbf{I}_{n,n} \Sigma$$

where $\underline{x}_n$ denotes a datapoint sampled from a Gaussian distribution with mean μ and covariance matrix Σ . Note that there is probably a mistake in question, intending to consider $\mathbb{E}[\underline{x}_n \underline{x}_n^T]$.

Either way, we will skip straight to the final question of interest, showing that

$$E[\Sigma_{ML}] = \frac{N-1}{N} \Sigma,$$

where $\Sigma_{ML} = \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{ML})(x_n - \mu_{ML})^T,$

$$\mu_{ML} = \frac{1}{N} \sum_{n=1}^N x_n.$$

To this end, we first note that

$$E[\mu_{ML}] = E\left[\frac{1}{N} \sum_{n=1}^N x_n\right] = \frac{1}{N} \sum_{n=1}^N E[x_n] = \mu.$$

Using i.i.d.

Now, $E[\Sigma_{ML}] = E\left[\frac{1}{N} \sum_{n=1}^N (x_n - \mu_{ML})(x_n - \mu_{ML})^T\right]$

$$= \frac{1}{N} \sum_{n=1}^N E[(x_n - \mu_{ML})(x_n - \mu_{ML})^T]$$

$$= \underbrace{\frac{1}{N} \sum_{n=1}^N E[x_n x_n^T]}_{(1)} - \underbrace{\frac{2}{N} \sum_{n=1}^N E[\mu_{ML} x_n^T]}_{(2)},$$

$$+ \underbrace{\frac{1}{N} \sum_{n=1}^N E[\mu_{ML} \mu_{ML}^T]}_{(3)}$$

Considering (1), it follows from the start of the exercise that

$$\frac{1}{N} \sum_{n=1}^N E[x_n x_n^T] = \frac{1}{N} \sum_{n=1}^N (\mu \mu^T + \underline{\Sigma}) = \mu \mu^T + \underline{\Sigma}$$

Considering (2), we have that

$$\begin{aligned} & - \frac{2}{N} \sum_{n=1}^N E[\mu_m x_n^T] \\ &= - \frac{2}{N} \sum_{n=1}^N E\left[\left(\frac{1}{N} \sum_{m=1}^N x_m\right) x_n^T\right] \\ &= - \frac{2}{N^2} \sum_{n=1}^N \sum_{m=1}^N E[x_m x_n^T] \end{aligned}$$

Note that when $m \neq n$, we assume $x_n \perp x_m$ and have $\text{Cor}(x_n, x_m) = \underline{0}$ (a matrix of zeros).

$$\begin{aligned} &= - \frac{2}{N^2} \sum_{n=1}^N \sum_{m=1}^N (\mu \mu^T + \mathbb{1}_{\{n=m\}} \underline{\Sigma}) \\ &= - \frac{2}{N^2} (N^2 \mu \mu^T + N \underline{\Sigma}) \\ &= -2 \left(\mu \mu^T + \frac{1}{N} \underline{\Sigma} \right) \end{aligned}$$

Finally, for (3) we have

$$\begin{aligned}\frac{1}{N} \sum_{n=1}^N \mathbb{E}[\mu_{nL} \mu_{nL}^T] &= \frac{1}{N} \sum_{n=1}^N \mathbb{E}\left[\left(\frac{1}{N} \sum_{i=1}^N x_i\right) \left(\frac{1}{N} \sum_{j=1}^N x_j\right)^T\right] \\&= \frac{1}{N^3} \sum_{n=1}^N \mathbb{E}\left[\left(\sum_{i=1}^N x_i\right) \left(\sum_{j=1}^N x_j\right)^T\right] \\&= \frac{1}{N^3} \sum_{n=1}^N \left(\sum_{i=1}^N \sum_{j=1}^N \mathbb{E}[x_i x_j^T]\right) \\&\stackrel{\text{Using same approach as for (2)}}{=} \frac{1}{N^3} \sum_{n=1}^N (N^2 \mu \mu^T + N \underline{\Sigma}) \\&= \mu \mu^T + \frac{1}{N} \underline{\Sigma}.\end{aligned}$$

Collectively, we now have

$$\begin{aligned}\mathbb{E}[\Sigma_{nL}] &= (\mu \mu^T + \underline{\Sigma}) - 2(\mu \mu^T + \frac{1}{N} \underline{\Sigma}) + (\mu \mu^T + \frac{1}{N} \underline{\Sigma}) \\&= \mu \mu^T + \underline{\Sigma} - (\mu \mu^T + \frac{1}{N} \underline{\Sigma}) \\&= \underline{\Sigma} - \frac{1}{N} \underline{\Sigma} \\&= \frac{N-1}{N} \underline{\Sigma}\end{aligned}$$

as we wanted.

Book exercise 2.6)

We are asked to find the mean, variance and mode (most likely value, i.e., maximum of the pdf) of the Beta-distribution.

To this end, note that for $\mu \sim \text{Beta}(\mu|a,b)$,

$$E[\mu] = \int_0^1 \mu \text{Beta}(\mu|a,b) d\mu$$

$$\stackrel{\text{def.}}{=} \int_0^1 \mu \cdot \left(\frac{\Gamma(a+b)}{\Gamma(a) \cdot \Gamma(b)} \mu^{a-1} (1-\mu)^{b-1} \right) d\mu$$

$$= \frac{\Gamma(a+b)}{\Gamma(a) \cdot \Gamma(b)} \int_0^1 \mu^{(a-1)+1} (1-\mu)^{b-1} d\mu$$

$$\stackrel{\nearrow}{=} \frac{\Gamma(a+b)}{\Gamma(a) \cdot \Gamma(b)} \cdot \frac{\Gamma(a+1) \Gamma(b)}{\Gamma(a+b+1)}$$

Using
(2.265)
with
 $a-1+1$

$$\stackrel{\nearrow}{=} \frac{\Gamma(a+b)}{\Gamma(a) \cdot \Gamma(b)} \cdot \frac{a \cdot \Gamma(a) \cdot \Gamma(b)}{(a+b) \cdot \Gamma(a+b)}$$

$$\stackrel{\nearrow}{=} \frac{a}{a+b}$$

Using $\Gamma(x+1) = x \Gamma(x)$.

Furthermore,

$$\begin{aligned} E[\mu^2] &= \int_0^1 \mu^2 B(\mu/a, b) d\mu \\ &= \int_0^1 \mu^2 \left(\frac{\Gamma(a+b)}{\Gamma(a) \Gamma(b)} \cdot \mu^{a-1} (1-\mu)^{b-1} \right) d\mu \\ &= \frac{\Gamma(a+b)}{\Gamma(a) \Gamma(b)} \int_0^1 \mu^{(a-2)+2} (1-\mu)^{b-1} d\mu \\ &\rightarrow \frac{\Gamma(a+b)}{\Gamma(a) \Gamma(b)} \cdot \frac{\Gamma(a+2) \Gamma(b)}{\Gamma(a+b+2)} \end{aligned}$$

Using
(2.265)

$$= \frac{(a+1)a}{(a+b+2)(a+b)}$$

Using $\Gamma(x+2) = (x+1)x \Gamma(x)$, i.e.,
 $\Gamma(x+1) = x \Gamma(x)$ gives,

It then follows that

$$\begin{aligned} \text{Var}(\mu) &= E[\mu^2] - E[\mu]^2 \\ &= \frac{(a+1)a}{(a+b+2)(a+b)} - \left(\frac{a}{a+b}\right)^2 \\ &= \frac{ab}{(a+b)^2(a+b+2)} \end{aligned}$$

Finally, we seek the maximum of $B(\mu|a,b)$ to find the mode.

$$\frac{\partial}{\partial \mu} (B(\mu|a,b)) = \frac{\partial}{\partial \mu} \left(\frac{\Gamma(a+b)}{\Gamma(a) \cdot \Gamma(b)} \cdot \mu^{a-1} (1-\mu)^{b-1} \right) d\mu$$

$$= \underbrace{\left(\frac{\Gamma(a+b)}{\Gamma(a) \cdot \Gamma(b)} \right)}_{(*)} \frac{\partial}{\partial \mu} (\mu^{a-1} (1-\mu)^{b-1}) d\mu$$

$$= (*) \cdot \left((a-1) \mu^{a-2} (1-\mu)^{b-1} - \mu^{a-1} (b-1) (1-\mu)^{b-2} \right)$$

$$= (*) \cdot \left(\mu^{a-2} (1-\mu)^{b-2} ((a-1)(1-\mu) - (b-1)\mu) \right)$$

$$= (*) \cdot ((a-1)(1-\mu) - (b-1)\mu)$$

$$= (*) \cdot ((a-1) - \mu(a+b-2))$$

Now, $\frac{\partial}{\partial \mu} \text{Beta}(\mu|a,b) = 0$ implies that

$$(*) \cdot (a-1) = (*) \cdot \mu(a+b-2)$$

$$\Rightarrow \mu = \frac{a-1}{a+b-2}$$

Additional exercise 1)

We consider $x \sim \text{Bern}(x/\mu)$ and want to derive the maximum likelihood estimator $\mu_{ML} = \frac{1}{N} \sum_{n=1}^N x_n$

from the log likelihood function

$$\ln(p(D/\mu)) = \sum_{n=1}^N \{x_n \ln(\mu) + (1-x_n) \ln(1-\mu)\}$$

To this end, we note that

$$\begin{aligned} \frac{\partial}{\partial \mu} (\ln(p(D/\mu))) &= \sum_{n=1}^N \left\{ x_n \frac{1}{\mu} + (1-x_n) \left(\frac{-1}{1-\mu} \right) \right\} \\ &= \frac{1}{\mu} \sum_{n=1}^N x_n - \frac{1}{1-\mu} \sum_{n=1}^N 1 + \frac{1}{1-\mu} \sum_{n=1}^N x_n \\ &= \frac{1}{\mu} \sum_{n=1}^N x_n - \frac{1}{1-\mu} N + \frac{1}{1-\mu} \sum_{n=1}^N x_n \\ &= \sum_{n=1}^N x_n \left(\frac{1}{\mu} + \frac{1}{1-\mu} \right) - \frac{1}{1-\mu} N \end{aligned}$$

From which it follows that

$$\frac{\partial}{\partial \mu} (\ln(p(D/\mu))) = 0$$

$$\Rightarrow \sum_{n=1}^N x_n \left(\frac{1}{\mu} + \frac{1}{1-\mu} \right) = \frac{N}{1-\mu}$$

$$\Rightarrow \frac{1}{\mu(1-\mu)} \sum_{n=1}^N x_n = \frac{N}{1-\mu} \Rightarrow \mu = \frac{1}{N} \sum_{n=1}^N x_n.$$

Additional exercise 2)

Let $\underline{x}$ follow a multinomial distribution with mean vector $\underline{\mu} = [\mu_1, \dots, \mu_K]$ for some $K \leq N$.

Note that we use one-hot-encoding such that $\sum_{k=1}^K x_k = 1$,

where x_k denotes the k 'th entry of $\underline{x}$ and

$$p(\underline{x} | \underline{\mu}) = \prod_{k=1}^K \mu_k^{x_k}$$

by assumption/construction.

Then

$$\sum_{\underline{x}} p(\underline{x} | \underline{\mu}) = \sum_{k=1}^K p(x_k = 1 | \underline{\mu})$$

Note that $\underline{x}$ can take K distinct values, characterized by each entry k being $= 1$, and all other zero.

$$= \mu_1 + \mu_2 + \dots + \mu_K$$

$$\stackrel{=}{=} 1$$

Must hold by
assumption/construction.

Else we do not have a distribution!

Furthermore, using indicator functions

$$\mathbb{1}_{\{k=j\}} = \begin{cases} 1, & \text{if } k=j, \\ 0, & \text{else,} \end{cases}$$

we have

$$\begin{aligned} E[\underline{x}] &\stackrel{\text{def.}}{=} \sum_{\underline{x}} \underline{x} p(\underline{x}|\underline{\mu}) \\ &= \sum_{k=1}^K \begin{pmatrix} \mathbb{1}_{\{k=1\}} \\ \mathbb{1}_{\{k=2\}} \\ \vdots \\ \mathbb{1}_{\{k=K\}} \end{pmatrix} \left(\prod_{k'=1}^K \mu_{k'}^{\mathbb{1}_{\{k'=k\}}} \right) \end{aligned}$$

A smart
way to describe $\underline{x}$
depending on k .

$$= \sum_{k=1}^K \begin{pmatrix} \mathbb{1}_{\{k=1\}} \\ \mathbb{1}_{\{k=2\}} \\ \vdots \\ \mathbb{1}_{\{k=K\}} \end{pmatrix} \mu_k$$

$$= \sum_{k=1}^K \begin{pmatrix} \mathbb{1}_{\{k=1\}} \mu_k \\ \mathbb{1}_{\{k=2\}} \mu_k \\ \vdots \\ \mathbb{1}_{\{k=K\}} \mu_k \end{pmatrix}$$

$$= \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_K \end{pmatrix}$$

$$= \underline{\mu}.$$